TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play outside in the park with her friends. One day, they found a big, round bird on the ground. Lily asked her friend, Sarah, “What’s wrong, I will make it breakfast,” her friend said.

Lily’s friend said, “I want to play with a silly bird!” Sarah gave Sarah a bird’s bag and took a big bite. Sarah was very happy and played with her toys all day long. When they reached, Lily felt better!

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once ·upon ·a ·time , ·there ·was ·a ·little ·g ir l ·named ·Lily . ·She
·lo ved ·to ·play · out s id e ·in ·the ·p ar k ·with ·her ·friend s .
·One ·day , ·they ·f ou nd ·a ·big , ·r ou nd ·b ir d ·on ·the ·g r ou
nd . ·Lily ·a s ked ·her ·friend , ·S ar a h , ·" W h at ’ s ·w r on g
, ·I ·w il l ·m a ke ·it ·b re a k f a st , " ·her ·friend ·said . ¶ L
ily ’ s ·friend ·said , ·" I ·want ·to ·play ·with ·a ·s il l y ·b ir d
! " ·S ar a h ·g a ve ·S ar a h ·a ·b ir d ’ s ·b a g ·and ·to o k ·a
·big ·b it e . ·S ar a h ·was ·very ·happy ·and ·play ed ·with ·her ·to y
s ·a ll ·day ·l on g . · W he n ·they · re a c he d , ·Lily ·f e l t
·b et t er !

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	3.8	53.2	22	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200, reached before the end-of-story token.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.